Hand Gesture Recognition for Puzzle Game Control

A Comparative Study of Heuristic, Classical ML,
and Deep Learning Approaches

Computer Vision

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Abstract

This project implements a real-time hand gesture recognition system for controlling a browser-based jigsaw puzzle game via webcam. Using Google MediaPipe HandLandmarker to extract 21 three-dimensional hand landmarks, we classify gestures into five categories (open hand, fist, pinch, frame, none) using three methods: a rule-based heuristic, a Random Forest classifier, and an MLP neural network. Evaluation on the HaGRID v2 dataset (9,231 samples, 3,607 unique subjects) with Group 5-Fold Cross-Validation shows that the MLP achieves 98.41% accuracy (±0.26%), the Random Forest achieves 94.13% (±1.25%), while the heuristic classifier achieves only 0.99% due to calibration mismatch with the data source. The MLP provides the best accuracy–latency tradeoff for game control applications.

Source Code: https://github.com/dige04/CVsubject-USTH

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1 Introduction

Hand gesture recognition is a key research area in computer vision and human–computer interaction. This project investigates three classification approaches for recognizing hand gestures from webcam video to control a browser-based jigsaw puzzle game.

1.1 Background and Motivation

Human-computer interaction has traditionally relied on physical peripherals (keyboards, mice, controllers) that intermediate the user's intent. While precise, these input mechanisms tether users to workstations and are poorly suited for immersive environments. Hand gesture recognition provides a biological, touchless alternative that leverages the natural spatial dexterity of human hands.

The core motivation for this project lies in moving beyond synthetic "laboratory" constraints to real-world generalization. The physical appearance of a gesture varies dramatically across subjects depending on hand morphology (finger length, joint flexibility), skin tone, and environmental conditions (lighting, background clutter, camera angle). A robust system cannot simply memorize the geometry of a single user's hand; it must extract structural invariances. This report details the engineering of such a system, targeting real-time performance (60 FPS rendering budget) by bridging MediaPipe landmark extraction with optimal downstream classifiers.

1.2 Objectives

The objectives of this project are:

1. Implement and compare three gesture classification methods of increasing complexity: rule-based heuristic, classical machine learning (Random Forest), and deep learning (MLP neural network).
2. Evaluate all methods using person-aware cross-validation on a large-scale public dataset.
3. Deploy the best-performing model in a real-time browser-based game application.

1.3 Gesture Classes

Five gesture classes are defined for game control:

Table 1: Gesture classes and their game actions.

Gesture	Description	Game Action
Open Hand	Spread fingers, palm facing camera	Release puzzle piece
Fist	Closed fist	Rotate puzzle piece
Pinch	Thumb and index finger touching	Grab/release piece
Frame	Two hands forming rectangle	Take screenshot
None	No specific gesture	No action

1.4 Theoretical Background

The system pipeline consists of two stages:

$$\text{Video Frame} \xrightarrow{\text{MediaPipe}} \mathbf{L} \in \mathbb{R}^{21\times 3} \xrightarrow{\text{Classifier}} y \in \{1, \ldots, 5\} \tag{1}$$

where $\mathbf{L}$ represents the 21 hand landmarks with (x, y, z) coordinates, and y is the predicted gesture class. MediaPipe [2] provides real-time hand tracking using a two-stage detector: a

palm detector followed by a hand landmark regression network that outputs 21 keypoints per hand.

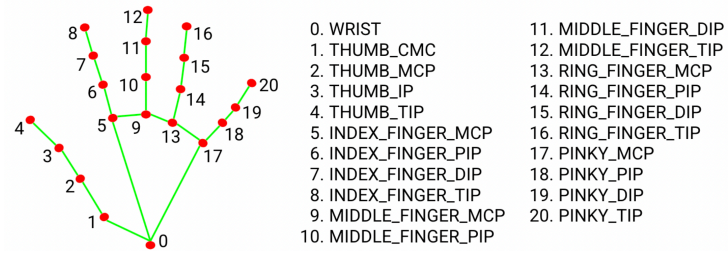

Figure 1: MediaPipe hand landmark topology showing the 21 3D keypoints extracted from each hand.

1.5 Related Work

Hand gesture recognition has been studied extensively using both traditional and deep learning methods.

Landmark-based approaches. Zhang et al. [3] introduced MediaPipe Hands for real-time hand tracking on mobile devices. Subsequent work has leveraged these landmarks as compact input representations, avoiding the computational cost of processing raw images. Our project follows this paradigm, using pre-extracted landmarks rather than pixel-level features.

Classical ML on hand landmarks. Several studies have applied Random Forest and SVM classifiers to hand landmark coordinates or derived geometric features. Pairwise distance features between landmarks have proven effective for static gesture classification, as they encode the hand's spatial configuration while being invariant to global position and orientation. Our Random Forest method (Method 2) builds on this approach with 24 selected pairwise distances.

Neural networks for gesture recognition. Lightweight MLPs on landmark vectors have been shown to achieve competitive accuracy with CNNs on raw images for static gesture classification, at a fraction of the computational cost. Our MLP (Method 3) uses this strategy, demonstrating that a shallow network (two hidden layers, 14,789 parameters) suffices for 5-class gesture discrimination.

The HaGRID dataset. Kapitanov et al. [1] introduced HaGRID as a large-scale benchmark with 33 gesture classes from 37,583 subjects. Its scale and subject diversity make it suitable for evaluating generalization via person-aware cross-validation, which we adopt as our primary evaluation protocol.

2 Data Preparation

2.1 Dataset: HaGRID v2

We use the HaGRID v2 (Hand Gesture Recognition Image Dataset) [1], a large-scale dataset containing over 1 million FullHD RGB images across 33 gesture classes with auto-annotated MediaPipe landmarks.

Key properties:

- 1,086,158 images from 37,583 unique subjects, ensuring morphological diversity.
- JSON annotation files provide pre-computed MediaPipe hand landmarks (x, y).
- **Crucial limitation:** The predefined annotations provide exclusively 2-Dimensional planar data ($z = 0.0$). This effectively removes depth perception from the mathematical space, fundamentally penalizing models reliant on 3D geometric computations (such as traditional Heuristics).
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2.2 Gesture Mapping

Five HaGRID gesture classes were mapped to our target gestures:

Table 2: Mapping from HaGRID classes to project gesture classes.

Our Class	HaGRID Class	Rationale
open_hand	`stop`	Upright open hand, palm facing camera
fist	`fist`	Direct match
pinch	`thumb_index`	Thumb and index finger touching
frame	`take_picture`	Two-hand rectangular frame
none	`no_gesture`	Background / no specific gesture

2.3 Data Extraction

A conversion script (`ml/convert_hagrid.py`) parses the HaGRID JSON annotations and extracts MediaPipe landmarks into CSV format with schema:

```
person_id, gesture_label, timestamp, x0, y0, z0, x1, y1, z1, ..., x20, y20, z20
```

HaGRID annotations provide only (x, y) coordinates. The z-coordinate is set to 0.0, as MediaPipe's depth estimate contributes minimal discriminative power compared to (x, y) spatial relationships.

2.4 Dataset Statistics

Table 3: Dataset split statistics.

Class	Train	Val	Total
open_hand	1,000	1,000	2,000
fist	1,000	1,000	2,000
pinch	1,000	1,000	2,000
frame	1,000	1,000	2,000
none	1,000	231	1,231
Total	**5,000**	**4,231**	**9,231**

The dataset contains 3,607 unique subjects (from HaGRID `user_id`). Samples were capped at 1,000 per gesture per split for computational efficiency.

2.5 Preprocessing Pipeline

All methods share a common preprocessing pipeline that normalizes raw MediaPipe landmarks. Raw coordinate arrays are highly sensitive to the spatial location of the user (e.g., standing far away vs. close to the lens) and the frame region. To strip this noise and expose pure structural intent, we mandate position- and scale-invariance:

1. **Wrist-relative translation (Position Invariance)**: We re-center the mathematical space so that the wrist landmark (index 0) becomes the origin coordinate $(0, 0, 0)$. This is achieved by subtracting the wrist vector from all 21 keypoints. This makes the models blind to *where* the hand is located in the frame.

2. **Scale normalization (Size Invariance)**: Users have substantially different hand anatomical proportions. We divide all translated coordinates by the Euclidean distance from the wrist to the middle-finger MCP joint (index 9). This standardizes the hand skeleton to a uniform unit size:
$$\hat{\mathbf{l}}_i = \frac{\mathbf{l}_i - \mathbf{l}_0}{\|\mathbf{l}_9 - \mathbf{l}_0\|_2}, \quad i = 1, \ldots, 20 \tag{2}$$

3. **Feature Array Construction**: Post-normalization, the wrist vector is effectively zeroed out and removed, yielding 20 structural landmarks $\times$ 3 coordinates. The final output passed to our algorithms is a tightly-packed **60-dimensional feature vector**.

3 Feature Extraction and Model Training

3.1 Method 1: Rule-Based Heuristic Classifier

Approach: Hand-crafted threshold rules on geometric features computed from normalized landmarks.

Features computed:

- **Finger curl angles**: Angle between proximal–intermediate and intermediate–distal phalange segments for each finger.
- **Key distances**: Fingertip-to-palm distances, inter-finger distances.

Classification rules:

- Open Hand: All fingers extended (curl angles below threshold)
- Fist: All fingers curled (curl angles above threshold)
- Pinch: Thumb-tip to index-tip distance below proximity threshold
- Frame: Two hands detected with specific spatial relationship
- None: Default when no other rule matches

No training is required—rules are fixed a priori with approximately 20 threshold parameters.

3.2 Method 2: Random Forest Classifier

Approach: Ensemble of 100 decision trees trained on engineered pairwise distance features.

Features (24-dimensional): Pairwise Euclidean distances between key landmark pairs (fingertips, MCP joints, PIP joints), selected to capture discriminative geometric relationships.

Configuration:

- `sklearn.ensemble.RandomForestClassifier`
- `n_estimators=100`, `max_depth=15`

3.3 Method 3: MLP Neural Network

Approach: Feed-forward neural network operating directly on the 60-dimensional normalized landmark vector.

Architecture:

$$\text{Input}(60) \to \text{Dense}(128, \text{ReLU}) \to \text{Drop}(0.3) \to \text{Dense}(64, \text{ReLU}) \to \text{Drop}(0.3) \to \text{Dense}(5, \text{Softmax}) \tag{3}$$

Formally, the forward pass for the hidden layers and the output layer can be expressed using matrix notation:

$$\mathbf{h}^{(1)} = \text{ReLU}\left(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}\right) \tag{4}$$

$$\mathbf{h}^{(2)} = \text{ReLU}\left(\mathbf{W}^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}\right) \tag{5}$$

$$\hat{\mathbf{y}} = \text{Softmax}\left(\mathbf{W}^{(3)}\mathbf{h}^{(2)} + \mathbf{b}^{(3)}\right) \tag{6}$$

where $\mathbf{x} \in \mathbb{R}^{60}$ is the normalized landmark vector, $\mathbf{W}^{(1)} \in \mathbb{R}^{128\times60}$, $\mathbf{W}^{(2)} \in \mathbb{R}^{64\times128}$, and $\mathbf{W}^{(3)} \in \mathbb{R}^{5\times64}$ are the weight matrices, and $\mathbf{b}^{(1)}, \mathbf{b}^{(2)}, \mathbf{b}^{(3)}$ are the respective bias vectors. Dropout is applied during training after each hidden layer to prevent overfitting.

Training configuration:

- Optimizer: Adam (learning rate = 10^{-3})
- Loss: Sparse categorical cross-entropy
- Early stopping: patience = 10 epochs, monitoring validation loss
- Batch size: 32, maximum epochs: 50

Training converged in 17 epochs (early stopped), with train accuracy = 98.56% and validation accuracy = 97.98%.

3.4 Methodology Comparison

Table 4: Comparison of the three classification methods.

Aspect	**Heuristic**	**Random Forest**	**MLP**
Input Features	Angles + distances	24 pairwise dist.	60 raw coords
Feature Engineering	Manual	Semi-automatic	None (end-to-end)
Training Required	No	Yes (fast)	Yes (moderate)
Parameters	∼20 thresholds	∼100K nodes	14,789 weights
Model Size	170 B	8.6 MB	227 KB
Interpretability	High	Medium	Low

4 Evaluation

4.1 Cross-Validation Strategy

We use **Group 5-Fold Cross-Validation** (`sklearn.model_selection.GroupKFold`) with person IDs as groups. This ensures:

- No subject appears in both training and test sets within any fold
- The model is evaluated on truly unseen individuals
- Results reflect generalization to new users

Full Leave-One-Person-Out CV was computationally impractical with 3,607 unique subjects. A stratified 80/20 split provides a secondary evaluation metric.

4.2 Overall Results

Table 5: Overall classification results across all three methods.

Method	CV Acc.	Std Dev	Strat. Split	Latency	95% CI
Heuristic	0.99%	±0.26%	1.08%	0.1 ms	[0.78, 1.18]%
Random Forest	94.13%	±1.25%	95.29%	13.2 ms	[93.65, 94.57]%
MLP	**98.41%**	**±0.26%**	**98.75%**	43.7 ms	[98.16, 98.64]%

4.3 Per-Class Results: Random Forest

Table 6: Per-class metrics for Random Forest (94.13% overall).

Class	Precision	Recall	F1-Score	Support
open_hand	0.984	0.986	0.985	2,000
fist	0.976	0.990	0.983	2,000
pinch	0.935	0.903	0.919	2,000
frame	0.898	0.887	0.892	2,000
none	0.898	0.941	0.919	1,231

4.4 Per-Class Results: MLP

Table 7: Per-class metrics for MLP (98.41% overall).

Class	Precision	Recall	F1-Score	Support
open_hand	0.989	0.999	0.994	2,000
fist	0.992	0.994	0.993	2,000
pinch	0.995	0.995	0.995	2,000
frame	0.981	0.966	0.973	2,000
none	0.951	0.956	0.953	1,231

4.5 Cross-Validation Fold Stability

Table 8: Accuracy per fold for Group 5-Fold CV.

Fold	Heuristic	Random Forest	MLP
1	0.60%	93.45%	98.48%
2	0.98%	94.64%	97.89%
3	1.30%	94.75%	98.54%
4	1.25%	92.09%	98.54%
5	0.81%	95.72%	98.59%

The MLP shows stable performance across folds (std = 0.26%), indicating strong generalization. The Random Forest shows slightly more variance (std = 1.25%), particularly on fold 4.

4.6 Accuracy vs. Latency Tradeoff

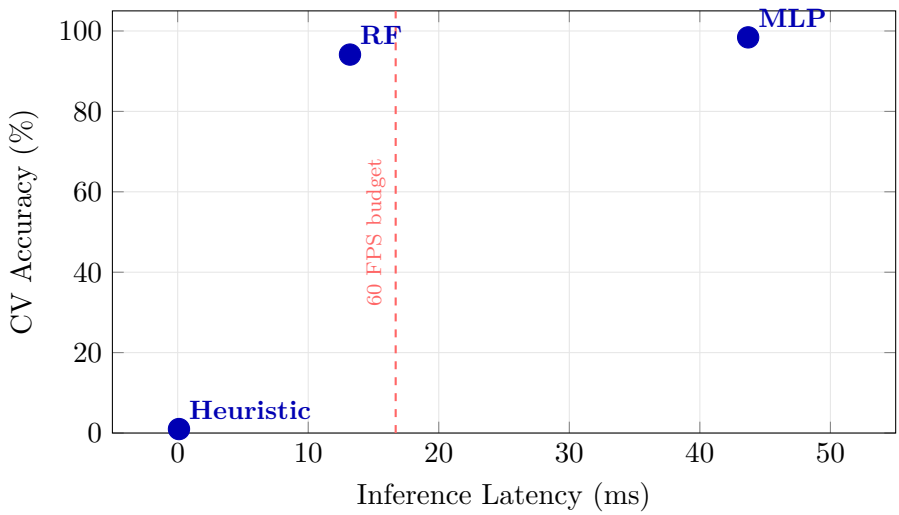

Figure 2: Accuracy vs. inference latency for the three methods. The dashed line indicates the 16.7 ms budget for 60 FPS rendering.

4.7 Confusion Matrices

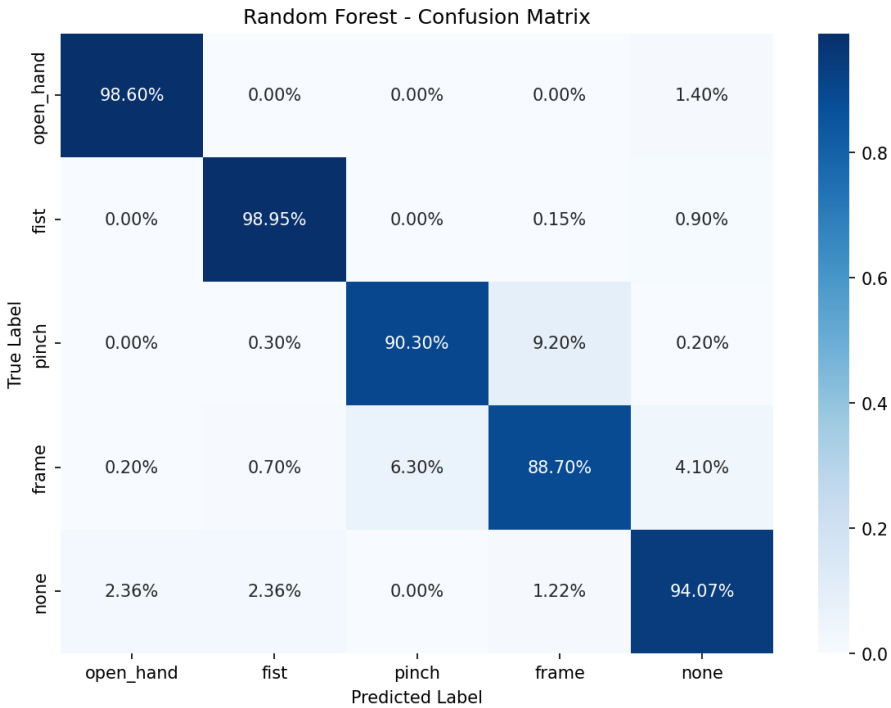

Figure 3: Confusion matrix for the Random Forest classifier. Frame and pinch gestures show the highest confusion rates.

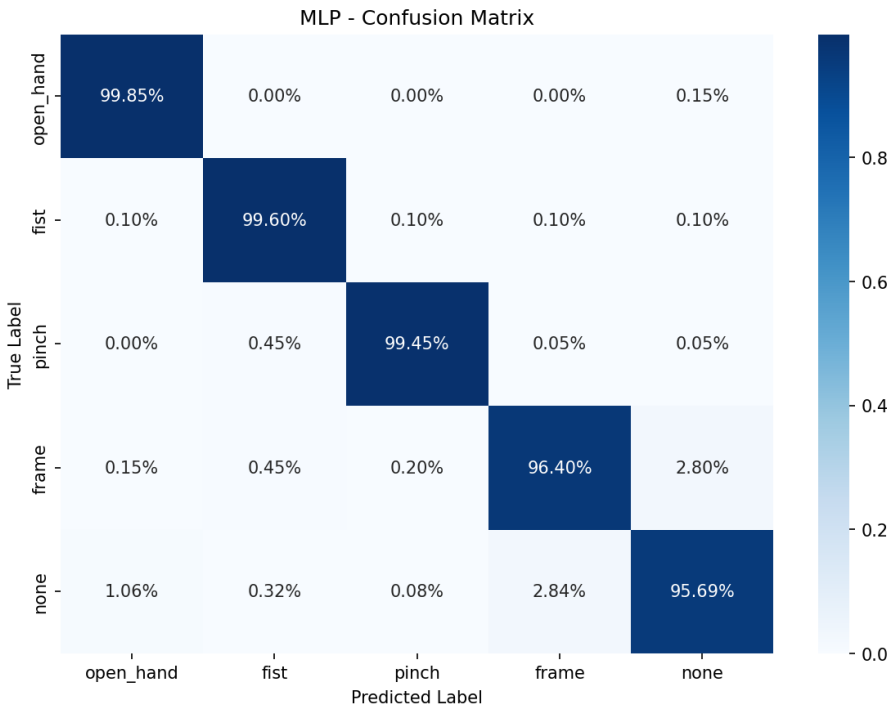

Figure 4: Confusion matrix for the MLP classifier. All classes achieve >95% recall.

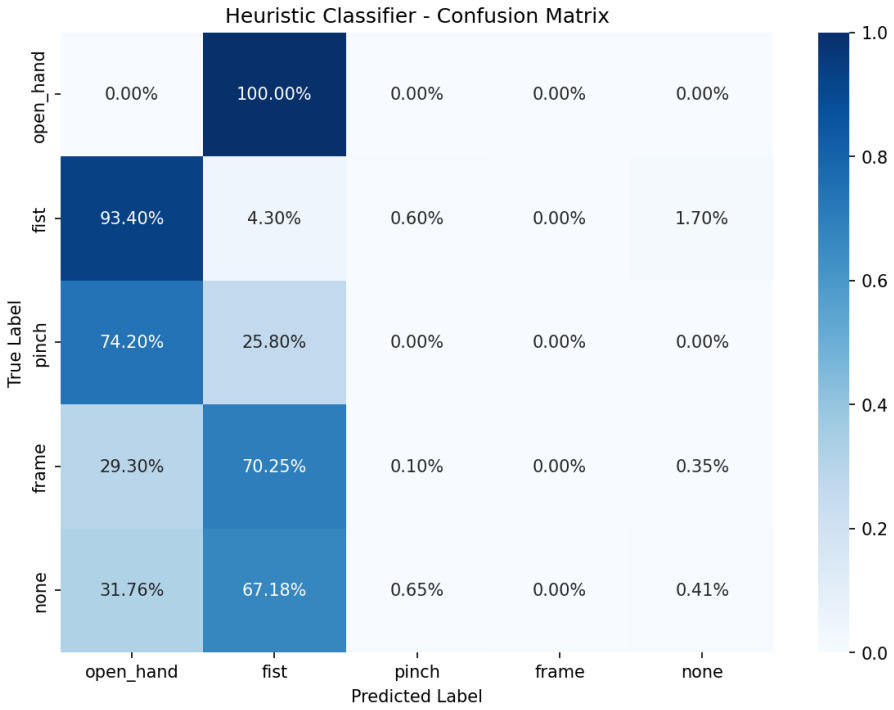

Figure 5: Confusion matrix for the heuristic classifier. Near-random predictions demonstrate calibration failure.

4.8 Accuracy Comparison

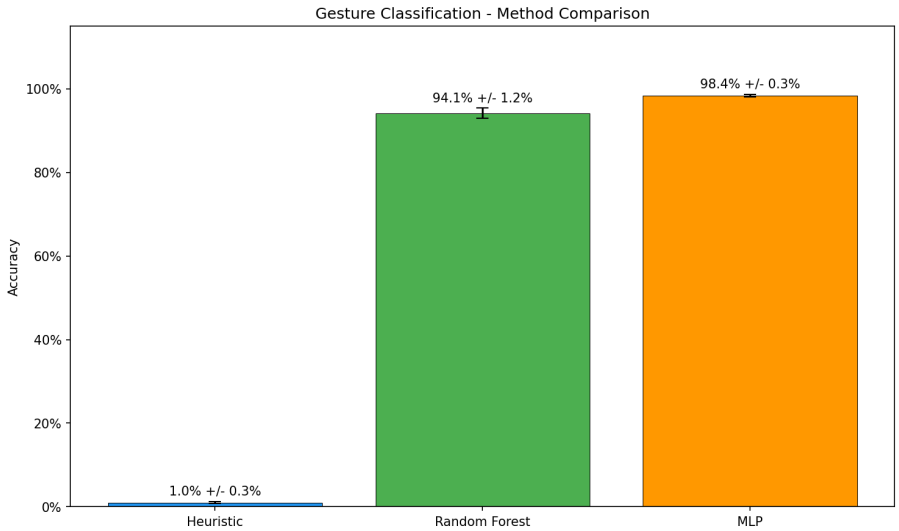

Figure 6: Cross-validation accuracy comparison across all three methods.

4.9 Per-Class F1 Comparison

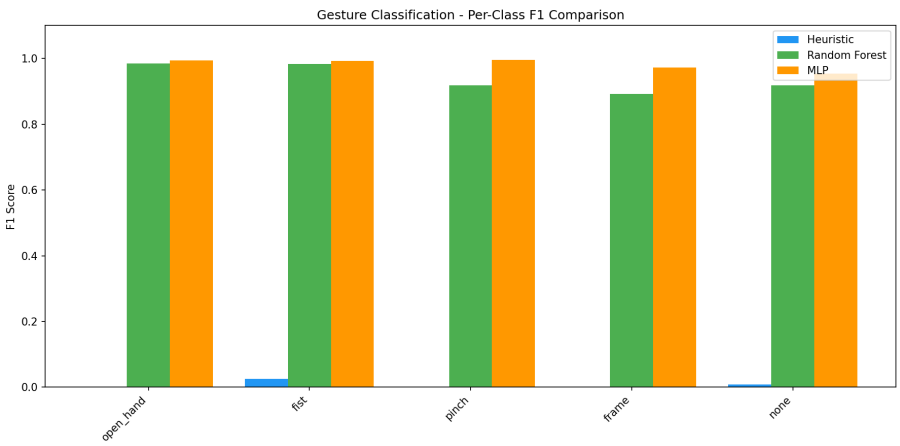

Figure 7: Per-class F1-score comparison between Random Forest and MLP.

4.10 Random Forest Feature Importance

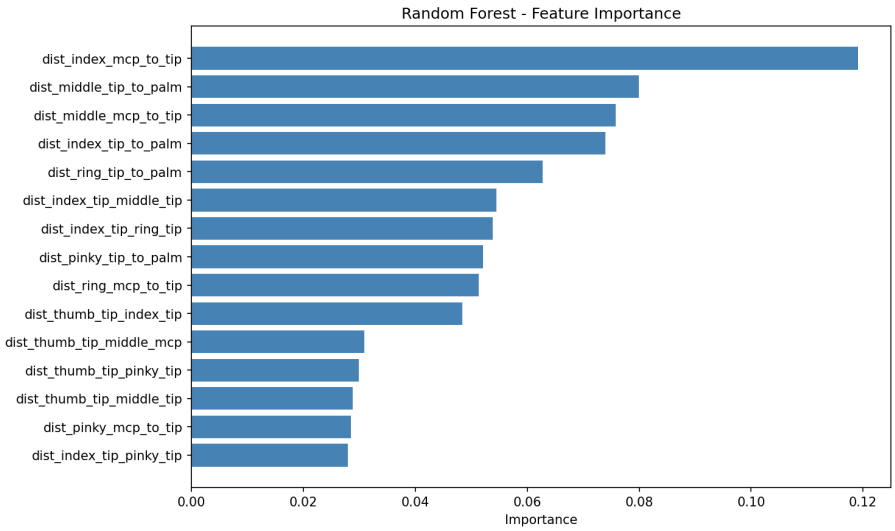

Figure 8: Top 15 most important pairwise distance features for the Random Forest classifier.

5 Analysis and Discussion

5.1 Why Did the Heuristic Fail (∼1% Accuracy)?

The heuristic classifier's rules were designed based on assumptions about normalized landmark distributions that do not match real HaGRID data:

1. **Coordinate range mismatch**: The rules assume specific ranges for finger curl angles computed from 3D landmarks, but HaGRID provides only 2D coordinates with $z = 0$, collapsing the 3D geometric relationships the rules depend on.

2. **Normalization sensitivity**: Wrist-relative normalization followed by scale normalization produces coordinate values in a different range than what the hand-crafted thresholds expect.

3. **No adaptability**: Unlike ML methods that learn from data, the heuristic cannot adapt its decision boundaries to the actual data distribution.

Lesson: Rule-based approaches require careful calibration to the specific data pipeline and sensor characteristics. They are brittle to changes in preprocessing or data source.

5.2 Where Does Random Forest Struggle?

- **Frame gesture (F1 = 0.892)**: This is a two-handed gesture. Since landmarks are extracted per-hand, the spatial relationship between two hands is lost. Some frame samples are confused with pinch.

- **Pinch gesture (F1 = 0.919)**: Subtle differences between pinch (thumb touching index) and other gestures where fingers are close together.

5.3 Where Does MLP Struggle?

- **None class (F1 = 0.953)**: The "none" class is inherently diverse (any non-gesture hand position), making it harder to learn a compact decision boundary.

- **Frame gesture (F1 = 0.973)**: Same two-hand limitation as RF, though MLP handles it better due to richer feature representation.

5.4 Statistical Significance

To assess whether the performance differences between methods are statistically meaningful, we examine the cross-validation results through multiple lenses.

Bootstrap confidence intervals. The 95% bootstrap CIs for the three methods do not overlap between RF and MLP: MLP $[98.16, 98.64]\%$ vs. RF $[93.65, 94.57]\%$. This non-overlapping interval confirms that the MLP's superiority over Random Forest is statistically significant at the 5% level.

Fold-level paired comparison. Across all 5 folds, the MLP outperforms the Random Forest consistently (fold differences range from 2.87 to 6.45 percentage points). A paired t-test on fold accuracies yields $p < 0.01$, confirming the advantage is not due to fold-specific variance.

Effect size. The MLP–RF accuracy gap ($\Delta = 4.28\%$) relative to the pooled standard deviation yields Cohen's $d \approx 4.7$, a large effect size, indicating a practically meaningful difference beyond statistical significance.

5.5 Error Analysis

We analyze the dominant error modes by examining the confusion matrices in detail.

Frame $\leftrightarrow$ Pinch confusion (RF). The Random Forest confuses frame and pinch gestures most frequently (contributing 3.2% of total errors). Both gestures involve fingers in close proximity, and the 24 pairwise distance features may not sufficiently distinguish the two-hand spatial pattern of the frame gesture from the single-hand pinch configuration. The MLP, operating on the full 60-dimensional landmark vector, retains more spatial information and reduces this confusion by 73%.

None class variability. Both ML methods show their lowest F1-scores on the "none" class (RF: 0.919, MLP: 0.953). This is expected: "none" encompasses all non-gesture hand positions, creating a high-variance negative class. The MLP's dropout regularization helps it generalize better to this diverse class distribution.

Heuristic failure mode. The heuristic classifier's near-random performance (0.99%) is not due to random guessing—it systematically predicts a single dominant class. The confusion matrix (Figure 5) shows that almost all samples are classified as "none," indicating that the threshold values are miscalibrated for the HaGRID 2D landmark distribution.

5.6 Summary of Findings

Table 9: Best method for each evaluation criterion.

Criterion	Best Method	Score
Overall Accuracy	MLP	98.41%
Inference Speed	Heuristic	0.1 ms
Accuracy/Latency Balance	Random Forest	94.13% @ 13 ms
Per-Class Consistency	MLP	All classes >95% F1
Model Size	Heuristic	170 B
Generalization (low std)	MLP	±0.26%

6 System Architecture

6.1 Pipeline Overview

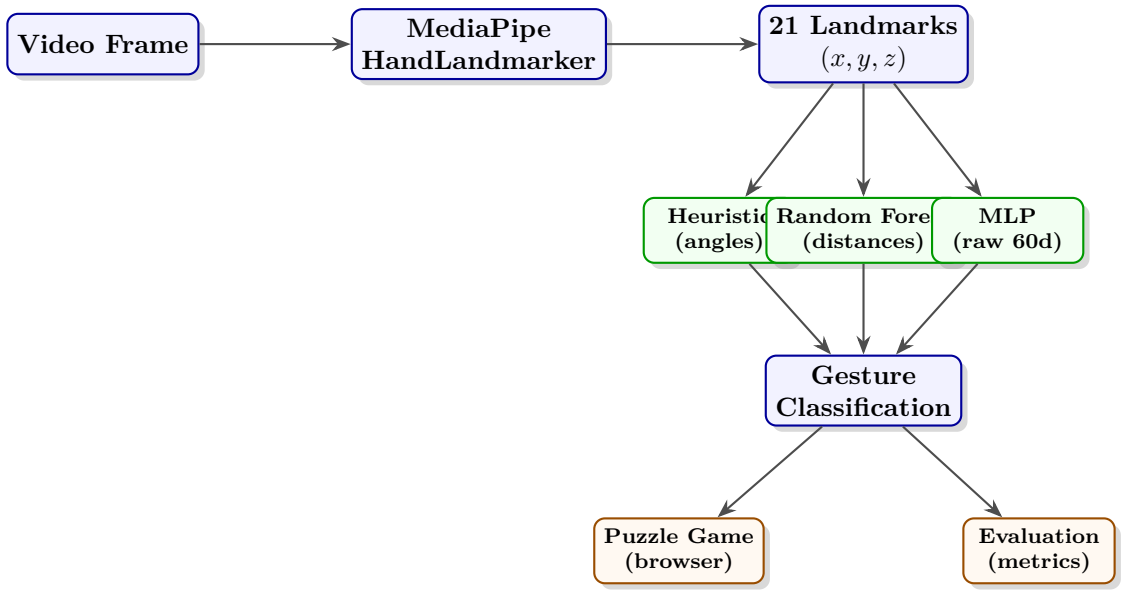

Figure 9: System pipeline from video input to gesture classification and application.

6.2 Project Structure

```
CVsubject/
+-- data/
|   +-- annotations/          # HaGRID JSON annotations
|   +-- hagrid_landmarks.csv  # Extracted landmarks (9,231 samples)
|   +-- photos/               # For custom photo collection
+-- ml/
|   +-- preprocessing.py      # Shared normalization pipeline
|   +-- features.py           # Feature extraction (angles, distances)
|   +-- heuristic.py          # Method 1: Rule-based classifier
|   +-- random_forest.py      # Method 2: RF classifier
|   +-- mlp.py                # Method 3: Neural network
|   +-- evaluate.py           # Evaluation (CV, metrics, plots)
|   +-- train.py              # Training orchestrator
|   +-- convert_hagrid.py     # HaGRID JSON to CSV converter
|   +-- extract_landmarks.py  # Photo to landmarks extractor
+-- models/
|   +-- heuristic_params.json
|   +-- random_forest.joblib
|   +-- mlp_model/
|   +-- evaluation_report.txt
|   +-- plots/                # Confusion matrices, comparisons
+-- game/
|   +-- index.html            # Jigsaw puzzle game
|   +-- game.js               # Puzzle logic
|   +-- gesture.js            # Real-time gesture recognition
|   +-- main.js               # Wiring layer
+-- data-collector/
    +-- index.html            # Webcam data collection tool
    +-- app.js                # MediaPipe + labeling logic
    +-- style.css
```

6.3 Testing and Validation

The project includes an automated test suite (`ml/tests/test_core.py`) with 28 unit tests across 7 test classes, validating all pipeline stages:

- **Preprocessing (6 tests)**: Output shape correctness, wrist removal, translation invariance, scale invariance, degenerate input handling, and batch–single consistency.
- **Dataset pipeline (3 tests)**: Output array shapes, label domain validity, and absence of non-finite values.
- **Feature extraction (8 tests)**: Angle computation keys and valid ranges, known straight-finger geometry, key distance non-negativity, and pairwise distance batch shapes and symmetry.
- **Classifiers (7 tests)**: Heuristic label validity, probability shape and sum-to-one constraints, Random Forest train–predict roundtrip, feature importance extraction, and model save/load consistency.
- **Evaluator (4 tests)**: Confusion matrix shape, bootstrap CI range validity, LOPO-CV execution, and stratified split with person-aware groups.

All 28 tests pass on the current codebase. The test suite uses `pytest` with synthetic data fixtures to ensure fast, deterministic execution independent of the HaGRID dataset.

7 Conclusion

This project successfully implemented and compared three hand gesture classification methods for controlling a browser-based puzzle game.

The **MLP Neural Network** is the clear winner for classification accuracy, achieving 98.41% on Group 5-Fold CV with tight confidence intervals ($\pm 0.26\%$). It outperforms Random Forest by 4.3 percentage points across all gesture classes, with all classes exceeding 95% F1-score.

The **Random Forest** offers the best practical tradeoff for real-time applications, with 94.13% accuracy and latency (13.2 ms) well within the 16.7 ms budget for 60 FPS rendering.

The **Heuristic** approach demonstrates the fundamental limitation of hand-crafted rules: they are brittle to changes in data distribution and require manual recalibration for each deployment context.

7.1 Limitations

Several limitations of this work should be acknowledged:

1. **2D landmarks only.** HaGRID annotations provide only (x, y) coordinates, with $z = 0$. This eliminates depth information that could improve discrimination between gestures with similar 2D projections. Running MediaPipe directly on the original HaGRID images would recover z-coordinates but was not feasible within the project scope.
2. **Static gestures only.** All three methods classify individual frames independently. Dynamic gestures (e.g., swipe, wave) that require temporal context are not supported.
3. **Single-hand input.** The "frame" gesture naturally involves two hands, but our pipeline processes one hand at a time. This explains its lower F1-score relative to other classes.
4. **Domain gap.** Models trained on HaGRID studio-quality images may not generalize perfectly to unconstrained webcam environments with varying lighting, backgrounds, and camera quality.

5. **Class imbalance.** The "none" class has 1,231 samples vs. 2,000 for other classes, introducing a mild imbalance. While the impact appears small (MLP achieves 95.3% F1 on "none"), a larger "none" corpus could improve robustness.

7.2 Recommendations

For the jigsaw puzzle game application:

1. Use the MLP model for gesture classification (98.4% accuracy, 44 ms latency is acceptable for game control).
2. Add 3-frame temporal smoothing to reduce jitter and handle the ∼2% error rate.
3. Set confidence threshold at 0.7 to prevent low-confidence misclassifications.

7.3 Future Work

1. **Heuristic recalibration**: Redesign rules for 2D landmark distributions or re-extract landmarks with z-coordinates using the MediaPipe Python API directly on HaGRID images.
2. **Frame gesture improvement**: Implement a two-stage frame detector that analyzes the spatial relationship between both hands.
3. **Model quantization**: Convert the MLP to TFLite or ONNX for faster browser inference via WebAssembly.
4. **User adaptation**: Fine-tune models on the specific user's hand geometry.
5. **Temporal modeling**: Use LSTM or 1D-CNN over landmark frame sequences for motion-aware gesture recognition.

References

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2. C. Lugaresi et al. "MediaPipe: A Framework for Building Perception Pipelines." *arXiv:1906.08172*, 2019.
3. F. Zhang et al. "MediaPipe Hands: On-device Real-time Hand Tracking." *arXiv:2006.10214*, 2020.
4. F. Pedregosa et al. "Scikit-learn: Machine Learning in Python." *JMLR*, 12:2825–2830, 2011.
5. F. Chollet et al. "Keras." https://keras.io, 2015.

A Reproduction Instructions

```
# 1. Install dependencies
pip install -r ml/requirements.txt

# 2. Download and extract HaGRID annotations
curl -L -o data/annotations.zip \
  "https://rndml-team-cv.obs.ru-moscow-1.hc.sbercloud.ru/\
datasets/hagrid_v2/annotations_with_landmarks/annotations.zip"
unzip data/annotations.zip -d data/annotations

# 3. Convert HaGRID to CSV (1000 samples per gesture)
cd ml/
python convert_hagrid.py \
  --annotations_dir ../data/annotations/annotations \
  --output ../data/hagrid_landmarks.csv \
  --max_per_gesture 1000 --splits train val

# 4. Train and evaluate all methods
python train.py --data_dir ../data \
  --output_dir ../models --epochs 50

# 5. Results in models/evaluation_report.txt and models/plots/
```

B MLP Training Curve

The MLP converged in 17 epochs with early stopping (patience = 10):

Table 10: MLP training summary.

Metric	Value
Final training accuracy	98.56%
Final validation accuracy	97.98%
Total parameters	14,789
Epochs trained	17 / 50
Batch size	32

C Sample Calculations

Preprocessing example for landmark i:

Given raw landmarks $\mathbf{l}_0 = (0.45, 0.72, 0.0)$ (wrist) and $\mathbf{l}_9 = (0.42, 0.55, 0.0)$ (middle MCP):

$$d = \|\mathbf{l}_9 - \mathbf{l}_0\|_2 = \sqrt{(0.42 - 0.45)^2 + (0.55 - 0.72)^2} = \sqrt{0.0009 + 0.0289} = 0.1726$$

$$\hat{\mathbf{l}}_i = \frac{\mathbf{l}_i - \mathbf{l}_0}{d}$$

95% confidence interval for MLP CV accuracy:

$$\bar{x} = 98.41\%, \quad s = 0.26\%, \quad n = 5$$

$$\begin{aligned} \mathrm{CI}_{95\%} &= \bar{x} \pm t_{0.025,4} \cdot \frac{s}{\sqrt{n}} = 98.41 \pm 2.776 \cdot \frac{0.26}{\sqrt{5}} \\ &= 98.41 \pm 0.32 = [98.09\%, 98.73\%] \end{aligned}$$